

Statistics Lab 52568 — Formula & Concept Sheet

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Sequence Basics (Lab 1)

Chargaff's rule: double-stranded DNA has $A \approx T$ and $C \approx G$.

Consequence: proportion plots of A&T track each other; so do C&G.

GC content of a bin of size b : $gc_k = (G_k + C_k)/b \in [0, 1]$

Key graphs: proportion vs. position (bin each base, plot count/ b); running median as trend line; scatter plots between base pairs (A vs T, C vs G should be most correlated).

Coverage & the Poisson Model (Lec 2–3)

Y_i = number of reads starting at base i , for $i = 1, \dots, I$.

J = total number of fragments in the experiment.

Average coverage:

$$\lambda = \frac{J}{I} = \frac{\text{total reads}}{\text{total bases}}, \quad \text{avg distance} = \frac{1}{\lambda}$$

Uniform Poisson — 3 assumptions:

1. Identical distribution: every fragment has the same p_i
 2. Uniformity: $p_i = 1/I$ for all i
 3. Independence: fragments land independently
- (1)+(3): $Y_i \sim \text{Binomial}(J, p_i)$. Adding (2) + Poisson approx (J large, $1/I$ small):

$$Y_i \sim \text{Poisson}(\lambda), \quad \mathbb{E}[Y] = \text{Var}[Y] = \lambda, \quad \text{SD}[Y] = \sqrt{\lambda}$$

Additive property: $X_1, \dots, X_K \sim \text{Pois}(\lambda)$ independent $\Rightarrow X_1 + \dots + X_K \sim \text{Pois}(K\lambda)$.

$\Rightarrow$ Bin of size b : $Y_k \sim \text{Pois}(\lambda_k)$, $\lambda_k = \sum_{i \in k} p_i \cdot J$.

VMR (Variance-to-Mean Ratio):

$$\text{VMR} = \frac{\text{Var}[Y]}{\mathbb{E}[Y]} = 1 \quad \text{under uniform Poisson}$$

Key graphs: histogram of Y_i vs fitted Poisson bars (log-scale y); coverage vs position with y-axis capped (mark outliers with different symbol); binned coverage full-chromosome plot.

Overdispersion & Variance Decomposition (Lec 3)

Real data: $\text{VMR} \approx 14.7$ — uniform model fails. Why?

Two-level model: $Y_k \sim \text{Pois}(\lambda_k)$, each bin has its own rate.

Law of Total Variance (exact identity):

$$\text{Var}(Y) = \underbrace{\mathbb{E}[\text{Var}(Y|\lambda)]}_{\text{avg within-group}} + \underbrace{\text{Var}(\mathbb{E}[Y|\lambda])}_{\text{variance of group means}}$$

Applied here:

$$\text{Var}(Y_k) = \underbrace{\mathbb{E}[\lambda_k]}_{\bar{\lambda} \text{ (Poisson noise)}} + \underbrace{\text{Var}(\lambda_k)}_{\text{rate heterogeneity}}$$

$$\text{VMR} = 1 + \frac{\text{Var}(\lambda_k)}{\bar{\lambda}}$$

$\text{VMR} = 1$ iff all λ_k equal. Spatial correlation of GC prevents $\text{Var}(\lambda_k)$ from averaging out under binning — rates in nearby bins are similar.

Robust estimators (resistant to outliers):

Center: **median**. Spread: **IQR**, **MAD** = $\text{med}|x_i - \text{med}(x)|$.

Use instead of mean/SD when data are heavy-tailed or have outliers.

Linear Regression for $f(\text{gc})$ (Lec 4)

Model: $Y_k = \beta_0 + \beta_1 \text{gc}_k + \epsilon_k$, $\mathbb{E}[\epsilon_k] = 0$

OLS estimate: $\hat{\beta}_1 = \widehat{\text{Cov}}(\text{gc}, Y) / \widehat{\text{Var}}(\text{gc})$

Residual plots: residuals vs. gc and vs. fitted values should be flat. Fan shape $\Rightarrow$ heteroscedasticity. CIs from `lm` are **wrong** under misspecification.

$$R^2 = 1 - \text{RSS}/\text{TSS}, \quad \text{RMSE} = \sqrt{\frac{1}{K} \sum_k r_k^2}$$

Global polynomials: sensitive to outliers — a few extreme points distort the entire fit.

Spline Regression (Lec 4)

Key idea: divide the predictor into regions at **knots** $t_1, \dots, t_m$; fit a polynomial in each piece; glue with smoothness constraints.

Progression: piecewise constant $\rightarrow$ piecewise linear (discontinuous) $\rightarrow$ continuous linear spline $\rightarrow$ cubic spline (continuous 1st & 2nd derivatives).

Cubic spline basis: $\hat{y} = b_0 + b_1x + b_2x^2 + b_3x^3 + \sum_j c_j(x - t_j)_+^3$, where $(u)_+ = \max(u, 0)$.

B-spline `bs(x, knots, degree=3)`: locally supported basis, same fit. More knots = more flexibility.

Heteroscedasticity check:

$$\text{SD ratio} = \frac{\text{SD}(\text{resid}, \text{gc} > \text{Q3})}{\text{SD}(\text{resid}, \text{gc} < \text{Q1})} \approx 2.25$$

SD ratio $\gg 1 \Rightarrow$ variance grows with gc/coverage.

Poisson GLM & Quasi-Poisson (Lab 5)

Poisson GLM: $\log \mathbb{E}[Y_k] = \beta_0 + \beta_1 \text{gc}_k$ (log link, counts).

Assumes $\text{Var}(Y) = \mathbb{E}[Y]$ — still violated ($\phi \approx 14$).

Quasi-Poisson: same mean, $\text{Var}(Y) = \phi \mathbb{E}[Y]$, ϕ estimated from data. More honest uncertainty.

Comparing models: R^2 for lm; McFadden pseudo- R^2 for GLM; $\tilde{R}^2 = 1 - \frac{\log L(\text{model})}{\log L(\text{null})}$.

Spearman r is rank-based — robust to outliers, but misleading when there is no real signal (centromere paradox: Spearman ≈ 0.95 despite $R^2 \approx 0$).

Bin size tradeoff: larger bins $\downarrow$ Poisson noise, $\uparrow$ GC signal blurring. Statistical elbow ~ 5 kb. Biological constraint: genes 10–100 kb.

Copy Number Estimation (Labs 6–8)

Model:

$$Y_k \sim \text{Pois}(\lambda_k), \quad \lambda_k = a_k \cdot f(\text{gc}_k) + \epsilon_k$$

Identifiability constraints: $\text{med}_k\{a_k\} = 1$, $\mathbb{E}_k[f(\text{gc}_k)] = 1$, $\mathbb{E}[\epsilon_k] = 0$.

Plug-in estimator:

$$\hat{a}_k = \frac{y_k}{\hat{f}(\text{gc}_k)} \cdot \hat{M}, \quad \hat{M} = \frac{1}{\text{med}_k \left\{ \frac{y_k}{\hat{f}(\text{gc}_k)} \right\}}$$

= divide out GC effect, renormalize so median = 1.

Variance of $\hat{a}_k$:

$$\text{Var}(\hat{a}_k) \approx \frac{\lambda_k + \sigma^2}{f(\text{gc}_k)^2} = \underbrace{\frac{1}{f(\text{gc}_k)}}_{\text{Poisson}} + \underbrace{\frac{\sigma^2}{f(\text{gc}_k)^2}}_{\epsilon \text{ noise}}$$

In our data: Poisson $\approx 16\%$, $\epsilon \approx 84\%$.

Noise models:

- Additive: $\lambda_k = a_k f + \epsilon$ — offset independent of signal
- Multiplicative: $\lambda_k = a_k f \cdot \eta$, $\eta \sim N(1, \gamma^2)$ — error scales with signal; more biologically plausible (PCR amplifies proportionally)

Two-sample estimator (Lec 8):

$$\hat{a}_k^{2\text{-sample}} = \frac{\hat{a}_k^{\text{tumor}}}{\hat{a}_k^{\text{healthy}}} = \frac{y_k^T / \hat{f}^T(\text{gc}_k)}{y_k^H / \hat{f}^H(\text{gc}_k)}$$

Fit a *separate* GC spline per sample. Ratio cancels shared GC bias and germline CNVs.

Key diagnostic plot: corrected ratio vs. position — should be flat at 1 under H_0 . If coloured by CNV call, the y-axis must be the *corrected* ratio, not raw reads.

Aggregative model (Lab 7): fit spline at fine bins (1 kb), predict coarse cells (10 kb) by summing sub-cell predictions. Captures local GC structure while controlling noise.

Hypothesis Testing & Multiple Testing (Lab 9)

Statistic: $S_k = \log_2(\hat{a}_k^{2\text{-sample}})$. Under H_0 : $S_k \approx 0$.

Negative controls: locally stable bins (window median $\leq \log_2 1.10$, SD $\leq \log_2 1.15$), filtered on the *ratio* so germline CNVs pass correctly.

Empirical p-value: $p_k = |\{j : |S_j| \geq |S_k|\}| / |\text{ctrl}|$. No distributional assumption.

Calibration: P-P plot of held-out CN=1 bin p-values should track the uniform diagonal.

Bonferroni (FWER): $\alpha' = \alpha/m$. Too conservative ($m \sim 10^5$).

Benjamini–Hochberg (FDR): reject $H_{(k)}$ for $k \leq k^* = \max\{k : p_{(k)} \leq k\alpha/m\}$.

Controls $\mathbb{E}[\text{FP}/\text{sig}] \leq \alpha$. Preferred: tolerates a small known FP fraction for much greater power.

Key Numbers

λ (50–60 Mb)	$\approx 0.036/\text{base}$
Binned VMR	≈ 14.7
GC-coverage r	≈ 0.84
Spline R^2 ceiling (5 kb)	≈ 0.75
SD ratio (high/low GC)	≈ 2.25
R^2 gain: G+C vs single base	≈ 0.17
Quasi-Poisson ϕ (after GC)	≈ 14
CN MAE from baseline	≈ 0.15
Var split (Pois / ϵ)	16% / 84%
r (tumor vs healthy, 2.5 kb)	≈ 0.10
Stat elbow (bin size)	~ 5 kb
Centromere gap	$\sim 120\text{--}130$ Mb